

1. Administrative & Study Identification

Full Study Title: A Phase I, Open-label, Multi-center Study to Evaluate the Safety, Tolerability, Dosimetry, and Preliminary Activity of [225Ac]Ac-ETN029 in Patients With Advanced DLL3-expressing Solid Tumors **Short Title/Acronym:** Phase I Study of [225Ac]Ac-ETN029 in Advanced DLL3-expressing Solid Tumors **Protocol Number:** CESP359A12101 **NCT Number:** NCT07006727 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [225Ac]Ac-ETN029 (Actinium-225 labelled DLL3-targeted radioligand therapy) and [111In]In-ETN029 (diagnostic imaging agent) **Phase of Development:** Phase I **Medical Monitor:** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, dosimetry, and preliminary efficacy of [225Ac]Ac-ETN029 and the safety and imaging properties of [111In]In-ETN029 in patients with locally advanced or metastatic DLL3-positive cancers. **Secondary Objectives:** To determine the Maximum Tolerated Dose (MTD) and/or Recommended Dose for Expansion (RDE), characterize the pharmacokinetic (PK) profile, and assess preliminary anti-tumor activity (e.g., ORR, PFS, OS).

2.2 Background & Rationale Delta-like canonical Notch ligand 3 (DLL3) is a cell surface protein that is highly expressed in neuroendocrine (NE) carcinomas (such as Small Cell Lung Cancer [SCLC], Neuroendocrine Prostate Cancer [NEPC], and Gastroenteropancreatic Neuroendocrine Carcinoma [GEP-NEC]) but has minimal expression in healthy adult tissues. Targeted radioligand therapy (RLT) using Actinium-225 (a potent alpha-emitter) provides a mechanism to deliver localized ionizing radiation directly to DLL3-expressing tumor cells, addressing a critical unmet medical need for patients who have progressed on standard platinum-based chemotherapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation and dose expansion phases) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: To be determined (Standard Phase I cohort sizing) **Number of Sites:** Multi-center

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Locally advanced, unresectable, or metastatic DLL3-expressing solid tumors including SCLC, Large Cell Neuroendocrine Carcinoma (LCNEC) of the lung, NEPC, or GEP-NEC. 3. Disease progression following, or intolerance to, at least 1 line of systemic therapy, including platinum-containing chemotherapy. 4. At least one measurable lesion (per RECIST 1.1) that shows $[^{111}\text{In}]\text{In-ETN029}$ uptake higher than surrounding tissues on SPECT/CT.

4.2 Exclusion Criteria 1. Absolute neutrophil count (ANC) $< 1.0 \times 10^9/\text{L}$, hemoglobin $< 9\text{ g/dL}$, or platelet count $< 75 \times 10^9/\text{L}$. 2. eGFR $< 60\text{ mL/min}$ calculated using the CKD-EPI 2021 formula. 3. QT interval corrected by Fridericia's formula (QTcF) $\geq 470\text{ msec}$. 4. Presence of leptomeningeal disease, symptomatic CNS metastases, or history/current interstitial lung disease (ILD) or pneumonitis $\geq$ Grade 2. 5. Any prior DLL3-targeted therapy (except for SCLC patients) and any prior radioligand therapy (except for NEPC patients).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dose escalation and expansion levels of $[^{225}\text{Ac}]\text{Ac-ETN029}$, preceded by a diagnostic and dosimetry imaging scan using $[^{111}\text{In}]\text{In-ETN029}$. **Route of Administration:** Parenteral (Intravenous) **Duration of Treatment:** Administered in protocol-defined cycles until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive oncology care. **Prohibited:** Concurrent investigational therapies, overlapping radiation or radioligand therapy, and other systemic anti-cancer therapies outside of the protocol parameters.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening: Informed Consent, Medical History, Safety Labs, Imaging with $[^{111}\text{In}]\text{In-ETN029}$ (SPECT/CT) to verify target uptake
2	0	Baseline: Vitals, First dose of $[^{225}\text{Ac}]\text{Ac-ETN029}$, Post-dose safety monitoring
3-5	Cycle Visits	Interim: Safety Labs, Dosimetry/PK blood draws, RECIST v1.1 Tumor Assessments, AE tracking
6	End of Treatment	End of Study: End of Treatment safety visit, Long-term survival follow-up, and subsequent anti-cancer therapy tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs), general Adverse Events (AEs), Serious Adverse Events (SAEs), preliminary tumor response, and optimal dosimetry profiles. **Measurement Method:** CTCAE grading scale for safety, RECIST v1.1 criteria for tumor imaging response, and central imaging review.

7.2 Secondary & Exploratory Endpoints Pharmacokinetic (PK) profile parameters of [225Ac]Ac-ETN029. Overall Response Rate (ORR), Duration of Response (DoR), and Progression-Free Survival (PFS). Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits, blood counts (monitoring for hematologic toxicity associated with radiation), and renal/hepatic lab parameters. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** A Dose Escalation Committee (DEC) will review all safety and DLT data before opening subsequent dose-level cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving at least one dose of study drug); Efficacy Evaluable Set (patients with measurable disease and at least one post-baseline assessment). **Sample Size Justification:** Guided by standard Phase I dose-escalation statistical parameters (e.g., Bayesian logistic regression model or 3+3 design) to adequately identify the MTD/RDE. **Handling of Missing Data:** Standard Phase I methodologies; safety data will not be imputed, and time-to-event efficacy endpoints will be right-censored at the last known evaluable date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of clinical data, source documents, and radiation safety protocols. **Audit Trail:** Data will be entered into a validated Electronic Data Capture (EDC) system utilizing strict eCRF locking mechanisms and data clarification form tracking.

11. Study Identifiers & Governance

Primary ID: CESP359A12101 **Registry Number:** NCT07006727 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Phase I Study of [225Ac]Ac-ETN029 in Patients With Advanced DLL3-expressing Solid Tumors **Official Regulatory Title:** A Phase I, Open-label, Multi-center Study to Evaluate the Safety, Tolerability, Dosimetry, and Preliminary Activity of [225Ac]Ac-ETN029 in Patients With Advanced DLL3-expressing Solid Tumors

12. Clinical Framework (Oncology/Targeted Therapy Specific)

Primary Condition: Advanced DLL3-expressing Solid Tumors (SCLC, LCNEC, NEPC, GEP-NEC) **Diagnosis Threshold:** Histological confirmation of target cancer and disease progression following standard lines of therapy (including platinum-based chemo). **Medical Methodology:** DLL3-Targeted Alpha-emitter Radioligand Therapy (RLT). **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) and Recommended Dose for Expansion (RDE).

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Phase I radioligand administration. **Education Protocol (HCP):** Site-level radiation safety training for handling/administration of alpha-emitting isotopes (Actinium-225). **Education Protocol (Patient):** Radiation safety precautions for home environments post-treatment and specific symptom-monitoring logs. **Quality Audit Mechanism:** Centralized imaging review for [111In]In-ETN029 uptake and RECIST 1.1 progression evaluations.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or withdrawal, followed by survival follow-up phase. **Onsite Visit Cadence:** Intensive monitoring during initial administration/dosimetry days (Cycle 1), relaxing to Day 1 of subsequent cycles. **Remote Monitoring Frequency:** Inter-cycle tele-health contacts for AE screening. **Checklist Review Interval:** Radiographic tumor assessments typically conducted every 8 to 12 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				